



CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

Dr. Prakash S Badal

Flow

- Announcement
- Recap
- Introduction

Announcement

- Lectures
 - M 8–8:50 am; T 1–1:50 pm; R 11–11:50 am; F 10–10:50 am
 - Fridays (tomorrow): no class
- Tutorial-1 on 3rd August (Monday) 3–5 pm
 - Venue: BSB 203 (same as the classroom)
- Check Moodle regularly (courses.iitm.ac.in)
 - *Yet to upload*
 - the course outline
 - notes from the first three lectures

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Course Flow

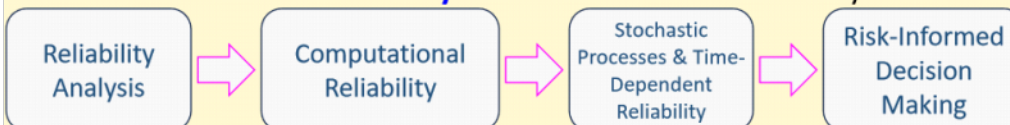
Module 1: **Probability** – Understanding Uncertainty



Module 2: **Statistics** – Learning from Data

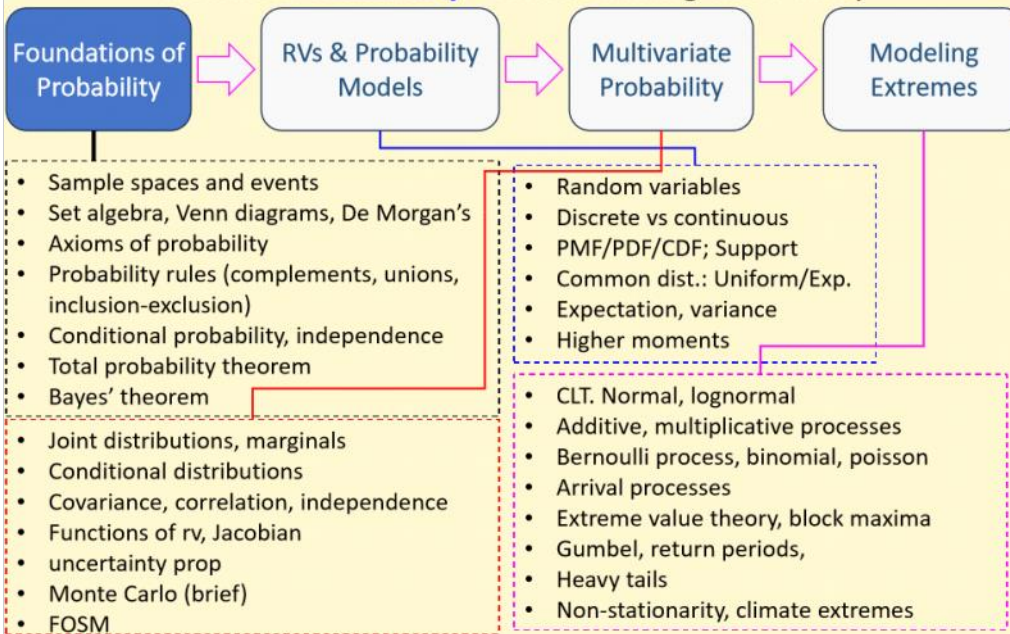


Module 3: **Reliability** – Decision under Uncertainty



Module I: Chapter 1 (M1C1)

Module 1: **Probability** – Understanding Uncertainty



Birthdays!

Birthday-Birthday

Jan <u>1</u> <u>16</u>	Feb	Mar <u>2</u> <u>14</u> <u>23</u>
Apr <u>13</u>	May <u>2</u> <u>14</u> <u>23</u>	Jun <u>7</u> <u>13</u> <u>16</u>
Jul <u>24</u> ✓✓	Aug	Sep
Oct	Nov	Dec

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Question 1

Probability that at least two people in this room share the birthday?



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Birthday Problem

- Intuition can mislead you.
- Birthday attack:
- $\Pr(\text{two people share birthday}) \gg$
 $\Pr(\text{A random person shares their b'day w/you})$
-

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Who is Reyansh?

Who is Reyansh?

- ~~Linda~~ ^{Riyansh} is 24 years old, **thoughtful, analytical**, and **exceptionally disciplined**
- He has a **B.Tech. in Mechanical Engineering** from IITM
- Outside academics, he enjoys **reading history** and **political philosophy**, closely follows **public affairs**, and frequently attends talks and panel discussions on **governance and public policy**

Which one the following is more likely.

OPTION A: Riyansh works for Tesla.

OPTION B: Riyansh works for Tesla & is active in public-policy discussion groups.

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Why Did We Get it Wrong?

Which events is more "PLAUSIBLE" X

— familiar story & stereotyping

Statistical Thinking: Ignore how convincing is the story.

ABR only $P(A \cap B) \leq P(A)?$

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Why Did We Get it Wrong?

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Bounds of $\Pr(A \cap B)$ ~~$\Pr(A \cup B)$~~ $= \Pr(A) + \Pr(B) - \Pr(A \cap B) \leq 1$

Lower Bound

$$\Rightarrow \Pr(A \cap B) \geq \Pr(A) + \Pr(B) - 1$$
$$\rightarrow 0 \leq \Pr(A \cap B)$$
$$\rightarrow \Pr(A) + \Pr(B) - 1 \leq \Pr(A \cap B)$$

Upper Bound

$$\Pr(A \cap B) \leq \min(\Pr(A), \Pr(B))$$

$$\max(0, \Pr(A) + \Pr(B) - 1) \leq \Pr(A \cap B) \leq \min(\Pr(A), \Pr(B))$$

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Boy-Girl Example

- A family has two children. You are told that

At least one child is a boy.

- What is the probability that both children are boys?



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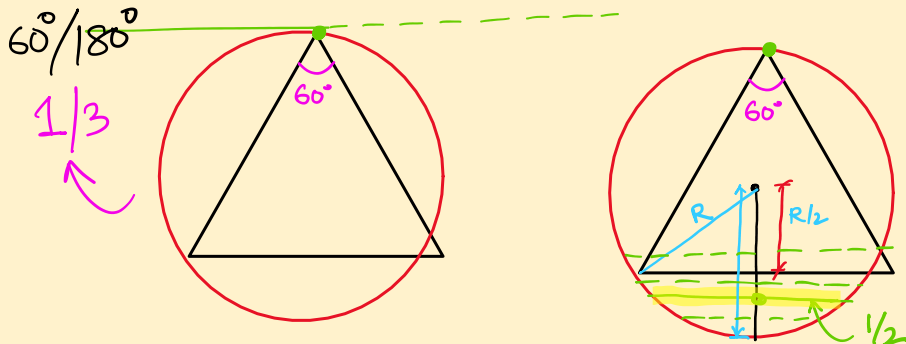
Boy-Girl Example

- *Equiprobability Bias*
 - When we have incomplete info
 - "Everything is equally likely"
- *"Monty Hall" Problem.*

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Bertrand Paradox

- Consider an **equilateral triangle** that is inscribed in a **circle**. Suppose a chord of the circle is chosen at random. What is the **probability that the chord is longer than a side of the triangle**?



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What about Probability Definition?

Probability,

I repeat an experiment n times.
An event A occurs n_A times

$$Pr(A) = \frac{n_A}{n} = \frac{\text{number of favorable outcomes}}{\text{number of trials}}$$

What if $n \rightarrow \infty$?

Not defined

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Probability Axioms

Mathematically, probability is defined using the following axioms:

- ① $0 \leq \Pr(A) \leq 1$ Non-negativity
- ② $\Pr(S) = 1$ Normalization
- ③ If $\underbrace{A \cap B = \emptyset}$, $\Pr(A \cup B) = \Pr(A) + \Pr(B)$
Additivity.

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Sets

→ Set: A collection of objects
 $\underbrace{\hspace{10em}}$ elements

→ $x \in A$ x belongs to A

$x \notin A$ x does NOT belong to A

$\mathcal{S}$ sample space:

→ Finite Set → Infinite set
e.g. Roll of a die, $\begin{matrix} + \text{ countably infinite} \\ + \text{ uncountable} \end{matrix}$

→ Prime Numbers.

→ $\Pr(S) = 1$

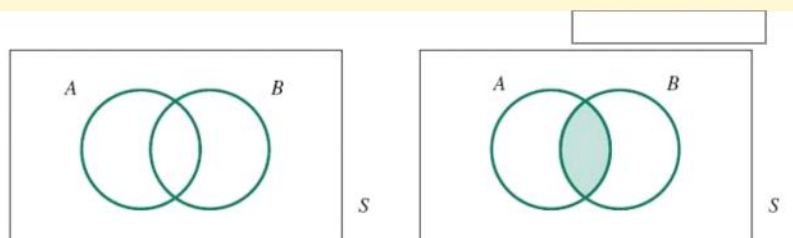
→ A^c : complement

→ all real numbers b/w 0 & 1

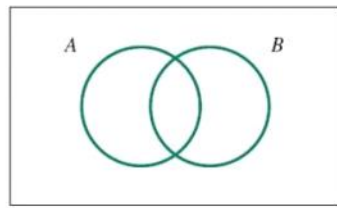
→ $\mathcal{S}^c = \emptyset$ Null Set

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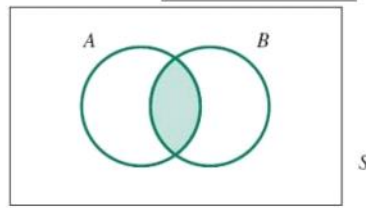
Venn Diagrams



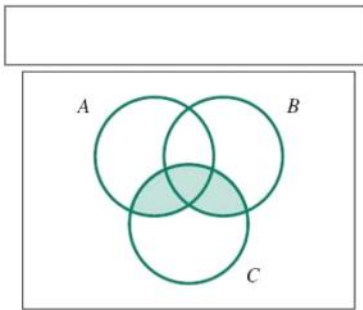
Venn Diagrams



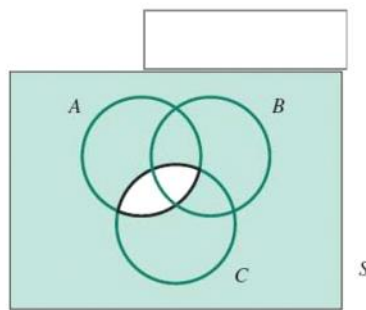
(a)



(b)

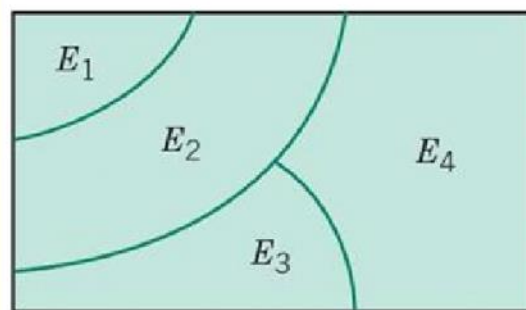


(c)

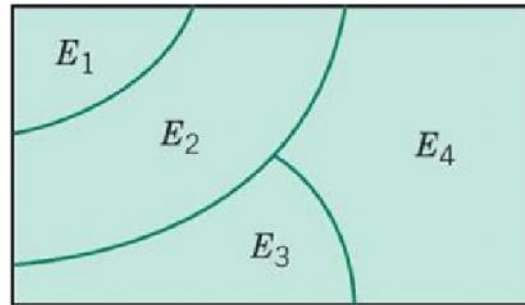


(d)

Set Theory: Mutually Exclusive Events



Set Theory: MECE



Set Theory: Other Rules

- Commutative rule

$$\begin{aligned}E_1 \cup E_2 &= E_2 \cup E_1 \\E_1 \cap E_2 &= E_2 \cap E_1\end{aligned}$$

- Associative rule

$$\begin{aligned}(E_1 \cup E_2) \cup E_3 &= E_1 \cup (E_2 \cup E_3) \\(E_1 \cap E_2) \cap E_3 &= E_1 \cap (E_2 \cap E_3)\end{aligned}$$

- Distributive rule

$$\begin{aligned}(E_1 \cup E_2) \cap E_3 &= (E_1 \cap E_3) \cup (E_2 \cap E_3) \\(E_1 \cap E_2) \cup E_3 &= (E_1 \cup E_3) \cap (E_2 \cup E_3)\end{aligned}$$

Set Theory: De Morgan's Rule

Rule-1

$$\overline{E_1 \cup E_2} = \overline{E_1} \cap \overline{E_2}$$

Prove it using a Venn Diagram

The rule is generalized to,

$$\overline{E_1 \cup E_2 \cup \dots \cup E_n} = \overline{E_1} \cap \overline{E_2} \cap \dots \cap \overline{E_n}$$

Rule-2

$$\overline{\overline{E_1} \cap \overline{E_2}} = \overline{\overline{E_1}} \cup \overline{\overline{E_2}}$$

Prove it using above equation

To prove this, apply previous rule $\overline{\overline{E_1}}$ and $\overline{\overline{E_2}}$:

$$\overline{\overline{E_1} \cap \overline{E_2}} = \overline{\overline{E_1}} \cup \overline{\overline{E_2}} = E_1 \cup E_2$$

Taking complement on both sides yields,

$$\overline{\overline{E_1} \cap \overline{E_2}} = \overline{E_1 \cup E_2}$$

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Probability rules

• Union: $\Pr(E_1 \cup E_2) =$

• Complement: $\Pr(\overline{E}) =$

• Inclusion-exclusion:

$$\Pr(E_1 \cup E_2 \cup E_3) =$$

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Probability rules: conditional prob.

- Conditional probability: $\Pr(E_1|E_2) =$

Probability of E_1 given that E_2 has occurred.

- Multiplication rule:
- Two events are called statistically independent, if

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Probability rules: multiplication rule

- Multiplication rule:

$$\Pr(E_1 E_2) = \Pr(E_1|E_2) \cdot \Pr(E_2)$$

- For n events,

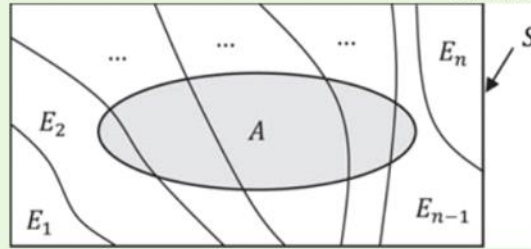
$$\Pr(E_1 E_2 \cdots E_n) =$$

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Probability rules: total probability rule

- If $E_1, E_2, \dots, E_n$ are n MECE events,

$$\Pr(A) =$$



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Probability rules: total probability rule

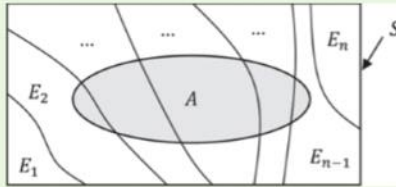
The probability that a building will sustain damage due to an earthquake depends on the intensity of shaking. Suppose the intensity is discretized into the states of weak, moderate, and strong. Assume it is determined that the probability of damage to the building is 0.01, 0.10, and 0.60 for the three levels of shaking intensity, respectively. Furthermore, based on the history of past earthquakes in the region, it is estimated that, for a randomly occurring earthquake, the probabilities for the three levels of shaking intensity at the site of the building are 0.90, 0.08, and 0.02, respectively. We are interested in the probability that the building will sustain damage during the next earthquake in the region.

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Probability rules: Bayes' rule

- If $E_1, E_2, \dots, E_n$ are n MECE events,



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Probability rules: Bayes' rule

- Assume in the previous example that the building has sustained damage. What can we learn about the intensity of the earthquake?

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Tutorial-1

- Topics
 - Probability
 - Venn diagrams
 - Conditional probability
- Questions: ~10-12 (question print-outs on the same day)
- Write your name, roll numbers on each sheet; bring staplers
- Closed-phone, closed-tablet, closed-AI
- Open-book, open-friends, open-TAs
- BYO
 - Empty/sheets, pen/pencil, calculators

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Thank you!